

Predictive Statistical Problems: Assignment Part 2

Slide 1 – Cover slide

Speaker 1:

Slide 2 – Introduction (1 minute)

"Good morning/afternoon, everyone. We are here today to present our project on predicting the likelihood of individuals experiencing health issues using machine learning. Our goal was to develop a predictive model that estimates health risks based on addictive behaviors and lifestyle factors."

"The motivation behind this project is clear: lifestyle choices significantly impact long-term health, contributing to approximately 60% of an individual's overall health and quality of life. Unhealthy habits like smoking, excessive alcohol consumption, poor diet, and a sedentary lifestyle are primary drivers of chronic conditions."

"We utilized a synthetic dataset of 3,000 individuals, which included data on smoking and drinking addiction patterns, alongside key lifestyle features such as age, exercise frequency, diet quality, hours of sleep, BMI, and mental health status. Our focus was on predicting a binary outcome: 'has_health_issues'."

Slide 3 – Methodology - Data Preparation (1 minute)

- **Title:** Methodology: Data Preparation
- **Key Points:**
 - **Dataset Overview:** Synthetic data from 3,000 individuals.
 - **Initial Cleaning:** Removed non-informative columns (ID, Name, City).
 - **Missing Values:** Handled missing data in education_level, social_support, and therapy_history.
 - **Data Types:** Ensured correct formatting for all variables.
- **Visual:** (Screenshot of the `df.isnull().sum()` output or a simplified table showing missing values and data types.)

Speaker 2

Slide 4 – (Methodology Part 1) (1 minute)

"Moving into our methodology, the first crucial step was data preparation. We started with a synthetic dataset containing 3,000 individual records. Our initial cleaning involved removing columns like id, name, and city which were non-informative for our predictive task."

"We then addressed missing values. As observed during our analysis, education_level, social_support, and therapy_history had a significant number of missing entries. For education_level, we imputed missing values using the mode, assuming the most frequent education level was a reasonable fill. For social_support and therapy_history, given their categorical nature and the potential impact of imputation, we opted to fill missing values with a designated 'Unknown' category to retain these records while clearly indicating the absence of information. Finally, we confirmed all data types were correctly formatted for subsequent analysis."

Slide 5 – Methodology - Exploratory Data Analysis (EDA) (1 minute)

- **Title:** Methodology: Exploratory Data Analysis (EDA)
- **Key Points:**
 - **Target Distribution:** Analyzed the balance of 'has_health_issues' (true/false).
 - **Numeric Feature Distributions:** Examined relationships between numerical features (e.g., age, BMI, smokes/drinks per day) and health issues.
 - **Categorical Feature Distributions:** Explored how health issues vary across categories (e.g., gender, mental health status, diet quality).
 - **Correlations:** Identified potential linear relationships between numerical variables.
- **Visual:** (Include two relevant plots from the EDA section of the notebook: e.g., "Distribution of Health Issues" bar plot and one or two compelling smokes_per_day or bmi distribution by health issues histograms. Or a heatmap of correlations).

Speaker 3

Slide 6 – Methodology Part 2 & Initial Findings (1 minute)

"Our next step was Exploratory Data Analysis, or EDA. This phase was critical for understanding the underlying patterns and relationships within our dataset."

"We began by examining the distribution of our target variable, `has_health_issues`. This allowed us to see how balanced our dataset was regarding individuals with and without health issues. We then delved into the numeric features, using histograms to visualize distributions for variables like age, BMI, and smoking/drinking habits, segmented by health status. This revealed insights into how these factors might influence health outcomes."

"Similarly, for categorical features such as gender, `mental_health_status`, and `diet_quality`, we used count plots to observe their distributions in relation to health issues. We also performed correlation analysis to identify any significant linear relationships between our numerical variables, which helped inform our feature selection for modeling. For instance, we observed a slightly higher number of individuals with health issues reporting higher children counts, and that individuals with health issues tended to report a somewhat higher number of attempts overall to quit drinking."

Slide 7 – Modeling and Results (1 minute)

- **Title:** Predictive Modeling & Key Results
- **Key Points:**
 - **Model Selection:** Logistic Regression and Classification Trees (briefly mention why – e.g., interpretability, suitability for binary classification).
 - **Feature Engineering:** Encoding categorical variables, scaling numerical features.
 - **Evaluation Metrics:** Accuracy, Precision, Recall, F1-Score, ROC AUC.
 - **Key Predictors:** Highlight 2-3 most influential features (e.g., `smokes_per_day`, `bmi`, `age`).
 - **Model Performance:** Summarize best model's performance (e.g., "Achieved [X]% accuracy with Logistic Regression and [Y]% with Classification Trees, with ROC AUC of [Z]").
- **Visual:** (A table summarizing the performance metrics for the selected models, or a visual showing feature importance from one of the models if available in the notebook.)

Speaker 4

Slide 8 – Results and Conclusion (2 minutes)

"Finally, we moved to the core of our project: predictive modeling. We experimented with two widely used machine learning algorithms for binary classification: Logistic Regression

and Classification Trees. Before training, we prepared our data by encoding categorical variables and scaling numerical features to ensure optimal model performance."

"We evaluated our models using a comprehensive set of metrics, including accuracy, precision, recall, and F1-score, with a particular focus on the ROC AUC score, which provides a robust measure of a classifier's performance across all classification thresholds."

"Our analysis revealed several key predictors for health issues. Among the most influential features were smokes_per_day, bmi, and age, consistently showing strong associations with the likelihood of having health issues. Our best performing model, [mention which one, e.g., Logistic Regression], achieved an accuracy of [X]% and a ROC AUC score of [Z], demonstrating its effectiveness in predicting health risks."

"In conclusion, our predictive model successfully identifies individuals at higher risk of health issues based on their lifestyle and addiction patterns. These insights can be invaluable for informing public health interventions and prevention strategies, allowing for more targeted and effective support for individuals."

Slide 9: Questions & Discussion

- **Title:** Questions & Discussion
- **Text:** "Thank you for your attention. We are now open for any questions."
- **Visual:** (Optional: a generic "Q&A" slide image.)